

Module 5

Editors, Projects & Archives

Learning objectives

Use a consistent src/tb/scripts/build layout

Find files and search with grep -R

Create and extract tar.gz archives

Back up projects before major changes

Prerequisites

Module 4 (basic scripts)

Learning path

Learning Path

Diagram: Learning Path
(render with mmdc when available)
flowchart LR
A[Editors] --> B[Project layout]
B --> C[find & grep]
C --> D[tar & zip]
D --> E[Backups]
E --> F[Module 6]

Course context

RTL projects mirror course repo structure

Exclude build/ from archives and Git

grep -R finds signals across a tree

Core topics

Self-check

```
$ ./scripts/module5.sh --check
```

Terminal: module_check

```

[0;36m=====
[0;36mModule 5: Environment self-check
[0;36m=====

[0;32m[OK] tar available
[0;32m[OK] zip available
[0;32m[OK] unzip available
[0;32m[OK] find available
[0;32m[OK] grep available
[0;32m[OK] module5 directory exists: /mnt/d/proj/universal-verification-methodology/learn_unix_
[0;32m[OK] project_structure/sample_project exists
[0;32m[OK] backup/backup_demo.sh exists

[0;32m[OK] All required checks passed.
```

Create archive

```
tar czf project_$(date +%Y%m%d).tar.gz project_dir/  
tar xzf project.tar.gz
```


Project structure

```
$ cd module5/examples/project_structure/sample_project && find . -type f | head -8
```

Terminal: project_tree

```
./build/.gitkeep  
./docs/notes.md  
./scripts/build.sh  
./src/main.v  
./tb/test_main.v
```

tar archives

```
cd module5/examples/tar_archives && tar czf /tmp/m5_demo.tar.gz demo_dir/ && tar tzf /tmp/m5_demo.tar.gz | head -
```

Terminal: tar_list

```
demo_dir/  
demo_dir/file1.txt  
demo_dir/file2.txt  
demo_dir/subdir/  
demo_dir/subdir/file3.txt
```

Practice

Exercise scaffold

```
./scripts/module5.sh --scaffold
```

Exercises

Scaffold `unix_course_project/`
`grep -R` in `sample_project`
Create a `tar.gz` backup

Summary & next steps

Organize src/tb/scripts/build

Archives for sharing

Next: Module 6 — Git basics

Next: Module 6: Git Basics for Course Work